

Learning that can show its work.

A local AI workspace connecting source evidence, misconception-aware practice and transfer checks.

The screenshot displays the TraceLearn application interface. On the left, a sidebar shows the 'YOUR WORKSPACE' with 'Database foundations' selected, and a 'LEARNING PATH' of 10 items, with '01 SQL NULL' highlighted. The main area is titled 'ORIGINAL COURSE - 10 CONCEPT PAIRS' and features a concept check exercise: 'Salaries are 100, 200, and NULL. Which rows survive WHERE salary <> 100?'. Below the question are four multiple-choice options: A (200 only), B (200 and NULL), C (All three rows), and D (No rows). A 'Check my understanding' button is at the bottom. To the right, a 'BACK TO THE SOURCE' panel shows 'ORIGINAL TEACHING NOTES' for 'WHERE keeps only TRUE', explaining SQL's three-valued logic and providing a table for the query results.

Workspace > Database foundations

Local AI ready 中文

YOUR WORKSPACE

Database foundations

+ Import your own notes

LEARNING PATH 10

01 SQL NULL

02 Aggregates

03 NOT IN

04 Candidate keys

05 Functional dependencies

06 Attribute closure

07 Second normal form

08 3NF vs BCNF

09 Decomposition

10 LEFT JOIN

Learning notebook 2

ORIGINAL COURSE - 10 CONCEPT PAIRS

From "I think" to "I understand."

Find the evidence. Spot the misconception. Test the idea again.

1 Diagnose 2 Trace evidence 3 Transfer & reflect

Practice Ask the source Notebook

01 CONCEPT CHECK

Salaries are 100, 200, and NULL. Which rows survive WHERE salary <> 100?

Choose one answer. A confident mistake is an opportunity to learn.

A 200 only

B 200 and NULL

C All three rows

D No rows

How sure are you? Unsure Somewhat Very sure

Check my understanding →

BACK TO THE SOURCE LOCAL

ORIGINAL TEACHING NOTES p. 1

WHERE keeps only TRUE

SQL uses TRUE, FALSE, and UNKNOWN in its three-valued logic. A WHERE clause keeps a row only when its predicate is TRUE; FALSE and UNKNOWN are both filtered out. For salaries 100, 200, and NULL, WHERE salary <> 100 returns only the row with 200. To include missing salaries, write salary <> 100 OR salary IS NULL.

SQL - THREE-VALUED LOGIC

salary <> 100

salary	predicate	WHERE
100	FALSE	excluded
200	TRUE	✓ kept
NULL	UNKNOWN	excluded

⌘ null-where

Source text is inspectable. AI interpretations still need judgment.

ML Empowerment Build Challenge 3.0 - Submission version 1.2

This report describes an implemented prototype and an internal technical benchmark. It does not claim improved grades, validated mastery, real-user adoption, or an independent classroom study.

1. Problem and product

Fluent explanations can make an idea feel familiar without demonstrating that a student can use it. TraceLearn connects an answer to its evidence, surfaces a possible misconception from a selected distractor, and asks a new question that applies the same concept. The initial scope is deliberately narrow: university database foundations.

The first worked scenario is SQL NULL. A learner predicts that salary $\neq 100$ keeps both 200 and NULL. The app checks the answer deterministically, explains that UNKNOWN is filtered by WHERE, opens the original passage, and then tests the same idea using a different condition.

The screenshot displays the TraceLearn application interface. On the left is a sidebar with a 'Database foundations' menu and a 'Learning path' list containing 10 items, with '01 SQL NULL' selected. The main area is titled 'From "I think" to "I understand."' and contains a 'CONCEPT CHECK' section. The question is: 'Salaries are 100, 200, and NULL. Which rows survive WHERE salary $\neq 100$?'. Below the question are four multiple-choice options: A (200 only), B (200 and NULL), C (All three rows), and D (No rows). Option B is selected. A feedback message states: 'A USEFUL GAP TO EXPLORE. Follow the evidence, then try again. The comparison with NULL is UNKNOWN. WHERE keeps only TRUE, so only 200 survives. You may be treating unknown as an ordinary value different from 100.' Below this, there are buttons for 'null-comparison' and 'null-where', and a link to 'Explore this reasoning with local AI'. At the bottom of the main area are buttons for 'Try the transfer check' and 'Retry'. On the right side of the interface, there is a 'BACK TO THE SOURCE' section titled 'WHERE keeps only TRUE'. It contains a paragraph explaining SQL's three-valued logic and a table showing the results of the WHERE clause for different salary values.

salary	predicate	WHERE
100	FALSE	excluded
200	TRUE	✓ kept
NULL	UNKNOWN	excluded

The screenshot shows a scripted software acceptance session, not a study participant.

Intended social value

Students can use their own material without uploading it to a cloud model or paying per API call. Downloads and adequate local hardware are still required. English/Chinese interface and source translations broaden accessibility; built-in questions retain English technical wording. Benefits for learning outcomes remain a hypothesis for a future study.

2. Implementation and boundaries

Layer	Implementation
Reading workspace	React + TypeScript; responsive source and practice panes; cancellation and recovery states.
Local service	Node.js + Express on 127.0.0.1; expected Host/Origin checks; no public deployment.
Documents and provenance	PDF.js or UTF-8 parsing; 10 MB / 50 PDF-page limits; source IDs, page/segment numbers, SHA-256.
Storage	SQLite stores source text, local vectors, questions and practice history. No local database is in the distributable.
Retrieval	0.75 cosine similarity + 0.25 query-token overlap; five highest-ranked passages.
Generation	Qwen3-4B via Ollama; Qwen3-Embedding-0.6B; temperature 0, seed 42, 8192 context, 1200 output-token cap, thinking disabled.
Checks	JSON schema; exact source-quote matching; answer echo/planning rejection; at most one retry.
Practice	20 original passages, 10 diagnostic/transfer pairs; deterministic grading. Imported-source questions are AI-assisted verbatim cloze exercises with a deterministic source-text answer check.

Imported practice was narrowed after a free-form generated item had four incorrect options. In the shipped path, the server selects a phrase from a real source sentence as the keyed answer; the local model supplies three distinct distractors. Completing the gap with the key reconstructs the source sentence exactly. This is source recall, not conceptual transfer. The app distinguishes citation location from entailment. A real quote can still be used to support a wrong claim; exact matching prevents invented quotations, not all hallucination. Prompt-injection resistance is bounded by the model and representative tests.

Key design tradeoff

A narrow, inspectable hybrid retriever reduces the context sent to a small local model. This can improve efficiency but can also omit necessary information. The full-source baseline receives all 20 original passages. Both conditions use the same final prompt and output checks, so this experiment compares context selection, not a different model or an unguarded baseline.

3. Local operation and reproducibility

All inference and data paths remain on the same computer. The browser talks to one local service; the service parses and stores sources, retrieves passages and asks the local model for structured output.

Path	Responsibility
Browser -> Node.js	Learning interface, local import, deterministic grading and report download.
Node.js -> SQLite	Source text, source hash, vectors, questions, attempt and chat records.
Node.js -> Qwen embedding	Create passage vectors once, then embed the current question.
Top five passages -> Qwen3	Produce a structured explanation with exact source excerpts.
Validation -> browser	Accept checked output or return an explicit failure after one retry.

Local runtime measurements

With both app and Ollama running under a macOS policy allowing only localhost outbound connections, real source-grounded answers succeeded. External DNS and direct-IP attempts were denied. A request after restarting Ollama took 6.62 seconds; the warm repeat took 4.17 seconds. Cached SQLite vectors were retained. Ollama reported approximately 6.23 GB allocated across the two loaded models; this is a model allocation snapshot, not peak whole-system memory.

Reproduce locally

```
npm run setup
npm start
http://127.0.0.1:4317
```

Model weights are not bundled. Exact model hashes, package lock, inference settings and all final outputs accompany the source. Tested on Apple M4 Pro with 24 GB memory; other operating systems were not tested.

4. Evaluation design

The following experiment was frozen on 14 September for version 1.0. Versions 1.1 and 1.2 add an evidence viewer, portable access and deterministic concept experiments. The answering pipeline and original course, raw results and benchmark scores remain unchanged.

The author-created dataset contains 80 fixed cases: 40 source-answerable questions, 20 source-unanswerable questions, and 20 multiple-choice questions with predetermined answers. Forty cases form the development split and 40 the holdout split. Related concepts occur in both splits; this is not a test of generalization to unseen courses.

The first pilot was stopped after a separate UI acceptance query returned a question echo with valid quotes. Two pilots exposed question echoes and option-letter leakage. The final v3 prompt requires completed explanations and only uses option-letter instructions for explicit multiple-choice questions. The dataset was not changed. Some pilot holdout requests had run, but their answer content was not used for tuning. Both pilot records and the final run are retained. This is an internal development benchmark, not an independently blinded evaluation.

Final holdout results

Metric	Full source baseline	Hybrid retrieval
Accepted answer on answerable cases	19/20	20/20
Structured insufficient flag	6/10	4/10
Correct extracted MCQ letter	7/10	9/10
Median wall time per request	6.25 s	3.21 s
Requests passing output checks	37/40	38/40

Accepted answer rate is not semantic correctness. MCQ correctness is based on extracted option letters; a formatting miss may count as incorrect even if prose implies the right answer. Expected-source overlap and keyword checks are diagnostic only and are not promoted as accuracy. Quote acceptance is conditioned on passing the validator. Wall time includes retrieval and retries, and can include local workload contention.

Agent-assisted reading of the complete holdout unanswerable outputs found prose refusals on 8/10 for each method, plus 2/10 validation failures. The structured flag undercounts textual refusals; both measures are retained.

Full run

The final run contains 160 requests across both methods. Raw JSONL includes accepted final responses, citations, retrieval passages, available retry records and timings; rejected calls retain error records. Dataset SHA-256:

e2eab651cddb82544f09a1493b77a059f184442151beab5fe606460e082217ef

See [evaluation/summary.json](#) and [evaluation/raw-results.jsonl](#) for split-level results and individual failures.

5. Interpretation, review and limitations

The product does not treat a correct practice answer as proof of durable understanding. Transfer checks are a concrete opportunity to apply a concept; they are not a validated psychometric instrument.

Agent-assisted review of all 20 source-answerable holdout cases per method judged 17/20 baseline and 18/20 grounded responses fully correct. These are non-independent judgments on an author-created, concept-overlapping dataset; they are not externally validated accuracy or learning gains.

Grounded in-4 used the correct SQL condition but wrote 10:00 instead of 100 in its explanation.

Grounded in-26 gave the formal 3NF definition but added an unsupported comparison to the informal slogan.

Baseline in-28 contradicted itself about BCNF implying 3NF; baseline in-36 contradicted its own source about dependency preservation.

Baseline in-40 failed validation and was not removed from the denominator. Exact quote matching did not prevent all semantic errors.

Structured insufficient flags undercounted prose refusals. On holdout unanswerable cases, both methods declined in prose on 8/10 and returned validation failures on 2/10. In development, grounded out-13 instead invented a university award policy; this is a real unsupported-answer failure.

What failed, and why it matters

The pilot exposed a substantive failure: quote validation alone accepted a response that repeated the question. Output validation was strengthened, but it cannot verify all meaning. The final result set preserves remaining rejected responses, unexpected refusals, answer mistakes, and formatting misses rather than dropping them from denominators.

This benchmark is small, written by the same development process as the course, and lacks independent human scoring. Source support review is agent-assisted and labeled accordingly. The baseline receives a small complete course; results may change for larger documents, other models, other languages or a different retrieval method.

Next research question

A prospective study could compare source-linked diagnostic practice with explanations alone using randomized conditions, independent question review, delayed transfer questions and consented participants. That study was not performed for this submission.

6. Verification and contribution

The final package includes executable unit tests, an isolated HTTP check, actual browser captures, an exported report, and local runtime evidence. SQL NULL examples execute against SQLite; selected candidate-key claims are checked by attribute closure. PDF import tests cover readable PDFs, damaged PDFs, blank/scanned pages and page limits.

Boundary	Evidence
Core implementation	Original v1.0: 17 unit tests passed. v1.1: 22 passed. v1.2: 48 automated tests passed, including HTTP integration; see the acceptance page.
HTTP routes	11 checks: hidden answers, grading, history, export, validation, imports, Origin and Host restrictions.
Browser	Actual diagnostic -> citation -> transfer -> notebook workflow; refresh restoration; desktop and narrow layouts; local AI and import paths.
Network independence	See evidence/offline-check.json for process-level localhost-only network validation and its scope.
Reproducibility	Locked package versions, frozen dataset and recorded model digests; exported source archive excludes personal runtime state.

AI contribution disclosure

Codex substantially assisted with design, code, original teaching and evaluation materials, testing, debugging, visual composition and documentation. Qwen provides local runtime responses. Agent-assisted checks are not represented as an independent human review. The public event encourages AI projects; separate organizer approval of AI-assisted code authorship has not been obtained.

Sources and licenses

Original app code and original course materials: MIT. Qwen models: Apache-2.0. Third-party dependencies retain their own licenses; exact package notices accompany the code. Model weights are downloaded separately.

Technical references: ollama.com/library/qwen3:4b; ollama.com/library/qwen3-embedding:0.6b; docs.ollama.com/api/embed; postgresql.org/docs/current/functions-comparison.html; postgresql.org/docs/current/functions-aggregate.html; postgresql.org/docs/current/queries-table-expressions.html.

Submission requirements and scoring: ml-build-challenge-3.devpost.com and [/rules](#). All screenshots show the actual application. Public source and downloadable materials: github.com/FENGJIA666/tracelearn (release v1.2.0). The application runs locally; no hosted inference service or live video is provided.

7. Try a claim, then change the conditions

Version 1.2 connects feedback to a Counterexample Lab. Predict which SQL rows survive, change the numbers or predicate, then compare the prediction with TRUE / FALSE / UNKNOWN. A second experiment traces attribute closure and removes each selected attribute to check minimality. These computations are deterministic; no language-model reasoning is used to decide the result.

TraceLearn

YOUR WORKSPACE

Database foundations

+ Import your own notes

LEARNING PATH 10

- 01 SQL NULL
- 02 Aggregates
- 03 NOT IN
- 04 Candidate keys
- 05 Functional dependencies
- 06 Attribute closure
- 07 Second normal form
- 08 3NF vs BCNF
- 09 Decomposition
- 10 LEFT JOIN

Learning notebook 2

Your notes. Your device.
Runs locally after model download.
No API key required.
[Local setup](#)

BUILD CHALLENGE 3.0 2026

LOCAL PROCESSING - NO ACCOUNT - NO API FEES

TraceLearn 1.2 - Follow the evidence.

100 200 NULL

Add row Remove last row

Condition value <> target Target number 100

WHERE value <> 100

2 Predict: which rows survive?

☐ Row 1 100 ☒ Row 2 200 ☒ Row 3 NULL

A counterexample: inspect the rows that differ.

Row / value	Your prediction	Predicate	WHERE
1 - 100	Exclude	FALSE	Excluded
2 - 200	Keep	TRUE	Kept
3 - NULL	Keep	UNKNOWN	Excluded Different

WHERE keeps only TRUE. FALSE and UNKNOWN are excluded. Change the condition to OR value IS NULL, predict again, and watch what happens to the missing value.

[Predict again with these inputs](#)

☐ null-comparison ☐ null-where

A bounded three-valued-logic simulator, differential-tested against real SQLite. It does not parse arbitrary SQL.

[Try a transfer question](#) [Export lab record \(6\)](#)

No AI is used in these computations. The latest 100 experiments stay in this browser; export to keep a copy. Predictions are separate from quiz scores and do not certify mastery.

SQL - THREE-VALUED LOGIC

salary <> 100

salary	predicate	WHERE
100	FALSE	excluded
200	TRUE	✓ kept
NULL	UNKNOWN	excluded

SQL - null-where

Source text is inspectable. AI interpretations still need judgment.

Browse passages

p.1 - WHERE keeps only TRUE

Source fingerprint

Actual v1.2 browser acceptance. The learner prediction is deliberately wrong: NULL is not an ordinary value different from 100. The result table exposes the mismatch. Switching to OR value IS NULL changes that row to TRUE. This is a scripted software check, not a student study.

One small file, a complete practice loop

The self-contained HTML is under 0.5 MB and includes 20 original questions, source notes, both editable experiments, practice and experiment exports, and all 160 recorded AI outputs. The quick tour opens real steps: predict, experiment, transfer, reflect. Full-app AI and imported documents still require Node.js, Ollama and downloaded models.

Experiment records retain inputs, predictions, source hash and time in browser storage. Export recomputes truth values and closures rather than trusting a saved score. Records are separate from quiz accuracy; clearing browser storage can erase them. Direct file:// opening remains unverified because browser automation blocks that scheme; the verified alternative is localhost static preview.

8. Version 1.2: verification and limits

TraceLearn.

YOUR WORKSPACE

Database foundations

+ Import your own notes

LEARNING PATH

10

01 SQL NULL

>

02 Aggregates

03 NOT IN

04 Candidate keys

05 Functional dependencies

06 Attribute closure

07 Second normal form

08 3NF vs BCNF

09 Decomposition

10 LEFT JOIN

📓 Learning notebook

2

🔌 Your notes. Your device.

Runs locally after model download.

No API key required.

Local setup

BUILD CHALLENGE 3.0

2026

☒ Candidate key: determines everything; no removable attribute

☐ Superkey, but not a candidate key: an attribute is removable

☐ Not a superkey: some attributes cannot be derived

🔍 The computation reveals a different result.

$\{A, D, B\}^+ = \{A, B, C, D\}$

1. Start with the selected attributes $\{A, D, B\}$

2. $B \rightarrow C$ adds $\{C\}$
 $\rightarrow \{A, B, C, D\}$

3. No rule can add another attribute. Stop.

This is a superkey, but not a candidate key. The closure covers the complete relation.

Try removing each attribute

Remove	Closure without it	Still a superkey?
A	$\{B, C, D\}$	No — needed
D	$\{A, B, C\}$	No — needed
B	$\{A, B, C, D\}$	Yes — removable

Closure is monotone: if removing any single attribute loses the superkey property, no smaller subset can be a superkey.

☐ Make another prediction

☐ closure

☐ keys

Results follow the stated dependencies. A small data sample cannot establish these constraints, and this trace does not automatically prove every normal form.

Try a transfer question

Export lab record (7)

NA 21 is used in these computations. The latest 100 experiments stay in this browser except to learn a minor. GradleLinux

Actual browser run: adding B to AD still determines the relation, but B is removable. The selected set is a superkey, not a candidate key. The trace explains the distinction without treating a single correct answer as proof of mastery.

Check	Observed result
SQL computation	1,600 row outcomes matched real SQLite; all Boolean three-valued truth-table entries checked separately.
Candidate keys	All 24 attribute subsets of the two built-in schemas matched an independent exhaustive two-tuple FD oracle; every proper subset checked for minimality.
Automated coverage	48 automated tests passed, including integration tests; 11 additional isolated HTTP checks passed. Both builds succeeded.
New local-model generation	Selected passage seven yielded two different quoted exercises in 2.956 s and 0.548 s; the next request returned 409 with no duplicate. Source hash and grading verified.
Installation identity	Two fixed copies used different ports; same-build relaunch reused its own server; changed source/build opened a new server without disturbing old processes.

Generation remains verbatim source recall: a distractor such as not null can mean almost the same thing as non-NULL, while only one option reconstructs the original quote. These items are not semantic-discrimination or transfer assessments. SQL experiments cover the shown numeric / NULL operators, not arbitrary SQL; key experiments use the stated dependencies, not constraints inferred from sample rows.

The implementation also cancels stale requests on navigation and opens notebook review as a fresh attempt. Controlled-delay browser tests use a clearly labeled mock API; local Qwen checks are recorded separately. Complete v1.2 procedures, raw acceptance records and remaining boundaries are in evidence/v1.2-acceptance.md. No newly tuned model benchmark, independent classroom outcome or contest ranking is claimed.

TraceLearn / Technical & Evaluation Report / 15 September 2026

9